Lishu Zhang

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Academic Appointment

2025–present Researcher, CREST, France

Education

2021–present Ph.D. in Economics, Tilburg University
Expected Completion: December 2026

2019–2021 Research Master in Economics, Tilburg University, with distinction

2017–2019 Master of Science in Economics, Stockholm School of Economics

2013–2017 Bachelor of Economics, Renmin University of China

Research Interests

Financial Economics, Environmental Economics, International Economics

Focus: Climate Finance, Cross-border Capital Flows, Production Networks, Critical Raw Materials

Working Papers

Measuring Critical Raw Material Dependence (w/ Peter Tankov and Olivier David Zerbib) ***(Job Market Paper)***

Firms can depend on critical raw materials they do not purchase directly because these materials are embodied in intermediate inputs several stages upstream. We develop a measure of firms' structural dependence on critical raw materials. Theoretically, we show that structural dependence can be represented by the sum across supply-chain paths of products of stage-level cost shares. The same statistic measures marginal-cost sensitivity to a material price shock and, under benchmark rationing, output sensitivity to a physical shortage. Empirically, we build material-specific production networks from engineering and trade evidence and calibrate the implied cost shares using physical input requirements, process yields, product prices, and sector markups. We map product exposure to firms using industry and segment information. The measure provides a framework for studying how material-specific disruptions affect firms along the value chain and whether structural dependence is reflected in equity prices.

Resistance and Arbitrage: International Trade in Brown Loans ***(Job Market Paper)***

Yihong Xia Best Paper Award, CICF 2025.

I develop a novel measure of carbon sensitivity in lending to assess reductions in portfolio exposure to brown assets. Using syndicated loan data, I show that countries with greater resistance to brown lending, proxied by economic development, experience faster shifts in the sectoral composition of loan portfolios. The decarbonization is driven primarily by domestic credit reallocation. I find consistent evidence of risk transfers to less regulated lenders and foreign countries, indicating arbitrage and incomplete regulations. Furthermore, lenders' climate risk-taking and transfer behaviors vary sharply by syndicate role, loan type, and specialization. The existence of international trade in brown loans has important implications for supervisory evaluation. Using the European Central Bank's climate guide, I show that accounting for regulatory leakage reveals effects contrary to common wisdom.

International ownership of brown shares and economic development (w/ Harald Benink, Harry Huizinga, and Louis Raes)

Using global share-ownership data for 2002–2021, we study how carbon-intensive firms come to be owned across countries at different levels of economic development and how this affects their environmental outcomes. Richer countries tilt away from carbon-intensive firms more than poorer countries do. Because shares must be held, market clearing leaves more of these firms in poorer-country hands. This poorer-country ownership predicts higher subsequent emissions and ESG incidents. We show that rich-country fossil-fuel divestment pushes firms toward such ownership, followed by worse environmental outcomes. Climate-motivated exit, by reallocating these firms toward less environmentally inclined owners, can work against the goal it serves.

When ESG information drives control: evidence from M&A deals

We study how ESG information affects the allocation of corporate control in mergers and acquisitions. Conditional on deal occurrence, firms with an ESG information advantage are more likely to acquire control, consistent with higher expected synergy. Environmental comparative advantage is a major determinant of M&A direction, beyond the governance gains emphasized in existing research. The effect is stronger in countries with higher institutional quality and richer information environments, and when markets have recently rewarded ESG-informed acquirers. Event-study results show asymmetric market perceptions of ESG information for acquirers and targets. ESG-informed control may also reduce within-deal uncertainty, as stock financing is less likely and less intensive in such deals. These findings suggest that used asset trade can reallocate control from less to more environmentally capable firms, facilitating the green transition.

Work in Progress

Localized Impacts of Transition Risks (w/ Thibaud Barreau, Jakob Kellermann, Peter Tankov, and Olivier David Zerbib)

Maps facility-level transition exposure and estimates local economic and environmental effects of realized shocks.

Making Climate Transitions Materially Feasible (w/ Peter Tankov and Olivier David Zerbib)

Studies how critical raw material constraints reshape least-cost transition pathways, technology choices, production, and trade.

Teaching Experience

Spring 2022–2024	Teaching Assistant, *Macroeconomics 3*, PhD level, Tilburg University
Fall 2021–2024	Teaching Assistant, *Economics of the European Union*, undergraduate level, Tilburg University
Fall 2021–2022	Teaching Assistant, *Economics of Banking and Finance*, undergraduate level, Tilburg University
Fall 2018	Teaching Assistant, *International Economics*, undergraduate level, Stockholm School of Economics

Grants, Scholarships, and Awards

2026	AFA Travel Grant, American Finance Association
2025	Yihong Xia Best Paper Award, CICF
2019–2021	Koopmans Scholarship, Tilburg University
2015	National Scholarship, Renmin University of China
2014	Outstanding School Cadre, Renmin University of China

Seminar & Conference Presentations

2026	AFA Poster Session; Junior Finance Research Day, CREST; HEC Paris Finance PhD Workshop (scheduled)
2025	CICF; AsianFA PhD Consortium; PhD-Economics Virtual Seminar
2024	FMA Annual Meeting; Canadian Economics Association Conference; DEARE Workshop, Utrecht University; ESSFM, Gerzensee; Imperial College PhD Conference in Economics and Finance; KU Leuven; Tilburg University (Finance)
2023	Tilburg University (Economics)

Professional Service

2021–2023	Member, Tilburg PhD Platform (TiPP)
2019–2020	Representative of Research Master's in Economics, Graduate Students' Society (GSS)

Computer Skills

STATA, R, MATLAB, Python, LaTeX.

Languages

English, French (intermediate), Mandarin

References

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